Christina M. Cavalluzzo

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EDUCATION

Boston University Boston, MA
Master of Science, Robotics and Autonomous Systems Aug 2025
Relevant courses: Medical Robotics, Soft Robotics, Mechatronics, Robot Motion Planning, Biological Feedback Control

Rensselaer Polytechnic Institute Troy, NY
Bachelor of Science, Mechanical Engineering | Studio Arts Minor May 2024
Honors: Dean's List 2020 – 2024, cum laude
Athletics and Leadership: NCAA Swim Team, Sprint Freestyle, 2020-2024

SKILLS

Mechanical Design: SolidWorks, Siemens NX, Fusion 360, AutoCAD, FEA, DFM/DFA, tolerance stackups, GD&T
Robotics & Controls: ROS2, inverse kinematics, robotic manipulation, motion control, EOAT design
Embedded & Compute: Python, MATLAB, Arduino, NVIDIA Jetson, Linux, C++
Certification: Fundamentals of Engineering Exam (Mechanical Engineering, August 2024)

EXPERIENCE

Amazon Robotics Westborough, MA
Robotics Hardware Development Engineer Co-op June 2025 – Feb 2026

- Co-led the end-to-end design and fabrication of a precision robotic end-of-arm tool (EOAT) enabling reliable manipulation of complex objects, iterating through CAD (SolidWorks) and SLA prototypes to validate tolerance, manufacturability, and kinematic range prior to final aluminum fabrication.
- Designed and prototyped interchangeable passive and sensorized soft robotic sleeves to enhance grip adaptability and compliance to enable safe manipulation of fragile objects, incorporating tactile and pressure sensing feedback.
- Incorporated two additional degrees of freedom to EOAT, increasing grasping versatility by over 80% while maintaining compactness and manufacturability.
- Integrated EOAT with Universal Robots (UR) arm, aligning mechanical mounts, configuring control interfaces, and implementing real-time control on NVIDIA Jetson and Arduino platforms (Python, ROS2, C++).
- Developed and optimized inverse kinematics for EOAT, improving positioning accuracy and response stability.
- Conducted bench and functional testing to evaluate grasp repeatability and actuator response.
- Designed multimodal sensor interfaces, optimizing placement to reduce cross-interference and improve awareness.

Paragon Energy Solutions Fort Worth, TX
Mechanical Engineer Intern May – Aug 2023

- Produced detailed engineering drawings (SolidWorks) for safety-critical, nuclear-grade mechanical components.
- Conducted structural and vibration analyses for safety critical components, supporting design verification activities and compliance with ASME and regulated engineering standards for nuclear safety protocols.
- Assisted in development and execution of structured test plans for mechanical and thermal validation.

BIOFerm Energy Systems Madison, WI
Mechanical Engineer Intern May – Aug 2022

- Produced AutoCAD fabrication models and system layouts for anaerobic digestion systems.
- Supported equipment specification and vendor selection using cost, manufacturability, and performance criteria.
- Evaluated supplier and partner options using technical, cost, and delivery performance metrics.

RESEARCH

Boston University Morphable Biorobotics Lab Boston, MA
Independent Researcher Oct 2024 – May 2025

- Integrated modular mechanical interface hardware for soft balloon actuators with jet propulsion robotic platform, iterating CAD (Fusion 360) and 3D-printed prototypes to achieve proper fit, sealing, and assembly reliability.
- Designed and fabricated bistable solenoid-magnet valve assembly from concept through working prototype to control multi-channel fluidic actuation, including magnetic force tuning and structural fixture design.
- Validated low-energy fluidic valve switching, achieving ~100-200 mJ switching energy and 5-7 ms actuation pulses.